Machine Learning Short Notes

A Beginner-Friendly Guide to Core Machine Learning Concepts
Based on the Machine Learning Specialization by Andrew Ng

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1. What is Machine Learning

- Machine Learning is a field of AI that allows computers to **learn from data without being explicitly programmed**.
- A program learns from **experience (E)** with respect to **task (T)** and **performance measure (P)** if its performance improves with experience.
- Two main types:
    - **Supervised Learning** – data has labels.
    - **Unsupervised Learning** – data has no labels.

Supervised Learning

2. Linear Regression Model

- Used to **predict continuous values**.
- Example: house price prediction.
- The model tries to fit a **straight line** through the data.
- Hypothesis function:

$h(x) = \theta_0 + \theta_1 x$

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3. Cost Function

- Measures **how good the model predictions are**.
- Linear regression usually uses **Mean Squared Error (MSE)**.

$J(\theta) = (1 / 2m) \Sigma (h(x) - y)^2$

Where

- m = number of training examples
- h(x) = predicted value
- y = actual value

Goal: **minimize the cost function**.

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4. Gradient Descent

- Optimization algorithm used to **minimize the cost function**.
- Updates parameters step-by-step.

$\theta = \theta - \alpha \, \partial J(\theta)$

Process:

1. Start with random parameters
2. Calculate cost
3. Update parameters
4. Repeat until minimum cost is reached

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5. Learning Rate

- Learning rate (α) controls **step size in gradient descent**.

Small α ; slow learning

Large α ; may overshoot minimum and diverge.

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6. Multiple Linear Regression

- Linear regression with **multiple input features**.

Example features:

- house size
- number of bedrooms
- number of floors
- age of house

Model:

$h(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \ldots + \theta_n x_n$

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7. Multiple Features

- Instead of one variable, model uses **feature vector**.

$x = [1, x_1, x_2, x_3 \ldots x_n]$

The **1 represents the intercept term**.

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8. Vectorization

- Vectorization allows ML computations to be done using **matrix operations instead of loops**.
- Makes algorithms **much faster and efficient**.
- Widely used in libraries like **NumPy and TensorFlow**.

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9. Feature Scaling

- Features may have **very different ranges**.

Example:

- house size = 2000
- bedrooms = 3

This slows gradient descent.

Feature scaling transforms features into similar ranges.

Example:

$x = (x - \mu) / s$

Where
μ = mean
s = range or standard deviation

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10. Feature Engineering

- Creating **new useful features from existing data**.

Example:

- frontage × depth → area
- age → age²

Help model **capture more patterns**.

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11. Polynomial Regression

- Extends linear regression by adding **power terms**.

Example:

$y = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

Allows model to **fit curves instead of straight lines**.

Classification

12. Logistic Regression

- Used for **classification problems**.
- Example:
    - spam detection
    - disease prediction

Instead of predicting numbers, it predicts **probabilities**.

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13. Decision Boundary

- A line (or curve) that **separates classes**.

Example:

- cancer vs non-cancer

Decision rule:

if $h(x) \geq 0.5 \rightarrow$ class 1
if $h(x) < 0.5 \rightarrow$ class 0

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14. Overfitting

- Model learns **training data too well**, including noise.
- Works well on training data but **poorly on new data**.

Causes:

- too many features
- complex models

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15. Regularization

- Technique to **reduce overfitting**.
- Adds penalty to large parameter values.

Regularized cost function:

$J(\theta) + \lambda \Sigma \theta^2$

λ = regularization parameter.

Neural Networks

16. What is Neural Network

- Inspired by the **human brain**.
- Consists of **layers of neurons**.
- Used for complex tasks like:
    - image recognition
    - speech recognition
    - NLP

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17. Neural Network Layers

Typical structure:

Input Layer → Hidden Layer(s) → Output Layer

- Input layer receives features
- Hidden layers learn patterns
- Output layer produces prediction

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18. Neurons and Layers

Each neuron:

1. Receives input
2. Computes weighted sum
3. Applies activation function
4. Produces output

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19. TensorFlow Implementation

TensorFlow is a popular **deep learning library**.

Example structure:

Sequential model
Dense layers
Activation functions

Used for:

- neural networks
- training models
- large scale ML systems

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20. Neural Network Implementation

Steps:

1. Define network architecture
2. Initializing weights
3. Perform forward propagation
4. Compute loss
5. Apply backpropagation
6. Update weights with gradient descent

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21. Activation Function

Activation functions introduce **non-linearity**.

Common ones:

- Sigmoid
- ReLU
- Softmax

Without activation functions, neural networks become **linear models**.

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22. Sigmoid Activation

Used for **binary classification**.

Function:

$g(z) = 1 / (1 + e^{-z})$

Output range:
0 to 1

Interpreted as **probability**.

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23. ReLU Activation

Rectified Linear Unit.

$ReLU(x) = max(0, x)$

Properties:

- very simple
- fast computation
- reduces vanishing gradient problem

Most common activation in hidden layers.

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24. Multiclass Classification

Used when there are **more than two classes**.

Example:

- classifying digits (0-9)
- object recognition

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25. Softmax

Softmax converts outputs into **probabilities for multiple classes**.

Formula:

$\text{softmax}(z_i) = e^{z^i} / \Sigma\, e^{z^j}$

Properties:

- probabilities sum to 1
- used in output layer.

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26. Backpropagation

Algorithms are used to **train neural networks**.

Process:

1. Calculate prediction
2. Compute error
3. Propagate error backwards
4. Update weights

This efficiently computes gradients.

Model Evaluation

27. Evaluating a Model

Common method:

Train / Validation / Test split.

Typical split:

- 60% training
- 20% validation
- 20% test

Training → learn parameters
Validation → tune hyperparameters
Test → final evaluation

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28. Bias and Variance

High Bias

- model too simple
- underfitting

High Variance

- model too complex
- overfitting

Goal: **balance bias and variance**.

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29. Regularization

Used again to **reduce variance and overfitting**.

Methods:

- L1 regularization
- L2 regularization

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30. Learning Curves

Graphs that show:

Training error vs dataset size
Validation error vs dataset size

Used to diagnose:

- bias
- variance problems

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31. Machine Learning Development Process

Typical workflow:

1. Define problem
2. Collect data
3. Train model
4. Evaluate model
5. Improve features
6. Tune hyperparameters
7. Deploy model

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32. Skewed Datasets

When one class is **much more frequent** than another.

Example:

- fraud detection
- disease detection

Accuracy becomes misleading.

Use:

- precision
- recall
- F1 score

Decision Trees

33. Decision Trees

A model that makes predictions by asking **a sequence of questions**.

Example:

Is age > 30?
Is income high?

Used for:

- classification
- regression

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34. Decision Tree Learning

Tree is built **top-down**.

Steps:

1. Select best feature
2. Split dataset
3. Repeat for subsets
4. Stop when condition met

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35. Tree Ensembles

Combining multiple trees improves performance.

Types:

- Random Forest
- Bagging
- Boosting

Advantages:

- higher accuracy
- reduced overfitting

Unsupervised Learning

36. What is Clustering

Clustering groups **similar data points together**.

Used when **labels are not available**.

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37. K-Means

Popular clustering algorithm.

Steps:

1. Choose K clusters
2. Initialize centroids
3. Assign points to nearest centroid
4. Update centroid mean
5. Repeat until convergence

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38. Anomaly Detection

Detects **unusual or rare events**.

Examples:

- fraud detection
- system failures

Uses probability models to detect anomalies.

Recommender Systems

39. Collaborative Filtering

Recommends items based on **user behavior**.

Example:

- Netflix movie recommendations
- Amazon product suggestions

Uses **user-item rating matrix**.

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40. Content-Based Filtering

Recommends items based on **item features and user preferences**.

Example:
If user likes:

- action movies

System recommends:

- similar action movies.

Dimensionality Reduction

41. Principal Component Analysis (PCA)

Used to **reduce number of features** while preserving information.

Benefits:

- faster computation
- easier visualization
- reduces noise

Steps:

1. Normalize features
2. Compute covariance matrix
3. Find eigenvectors
4. Project data to lower dimension